

Show that the procedure RANDOM-SAMPLE on the previous page returns a random m -subset S of $\{1, 2, 3, \dots, n\}$, in which each m -subset is equally likely, while making only m calls to RANDOM.

★ 5.4 Probabilistic analysis and further uses of indicator random variables

This advanced section further illustrates probabilistic analysis by way of four examples. The first determines the probability that in a room of k people, two of them share the same birthday. The second example examines what happens when randomly tossing balls into bins. The third investigates “streaks” of consecutive heads when flipping coins. The final example analyzes a variant of the hiring problem in which you have to make decisions without actually interviewing all the candidates.

5.4.1 The birthday paradox

Our first example is the *birthday paradox*. How many people must there be in a room before there is a 50% chance that two of them were born on the same day of the year? The answer is surprisingly few. The paradox is that it is in fact far fewer than the number of days in a year, or even half the number of days in a year, as we shall see.

To answer this question, we index the people in the room with the integers $1, 2, \dots, k$, where k is the number of people in the room. We ignore the issue of leap years and assume that all years have $n = 365$ days. For $i = 1, 2, \dots, k$, let b_i be the day of the year on which person i 's birthday falls, where $1 \leq b_i \leq n$. We also assume that birthdays are uniformly distributed across the n days of the year, so that $\Pr \{b_i = r\} = 1/n$ for $i = 1, 2, \dots, k$ and $r = 1, 2, \dots, n$.

The probability that two given people, say i and j , have matching birthdays depends on whether the random selection of birthdays is independent. We assume from now on that birthdays are independent, so that the probability that i 's birthday and j 's birthday both fall on day r is

$$\begin{aligned}\Pr\{b_i = r \text{ and } b_j = r\} &= \Pr\{b_i = r\} \Pr\{b_j = r\} \\ &= \frac{1}{n^2} .\end{aligned}$$

Thus, the probability that they both fall on the same day is

$$\begin{aligned}\Pr\{b_i = b_j\} &= \sum_{r=1}^n \Pr\{b_i = r \text{ and } b_j = r\} \\ &= \sum_{r=1}^n \frac{1}{n^2} \\ &= \frac{1}{n} .\end{aligned}\tag{5.7}$$

More intuitively, once b_i is chosen, the probability that b_j is chosen to be the same day is $1/n$. As long as the birthdays are independent, the probability that i and j have the same birthday is the same as the probability that the birthday of one of them falls on a given day.

We can analyze the probability of at least 2 out of k people having matching birthdays by looking at the complementary event. The probability that at least two of the birthdays match is 1 minus the probability that all the birthdays are different. The event B_k that k people have distinct birthdays is

$$B_k = \bigcap_{i=1}^k A_i ,$$

where A_i is the event that person i 's birthday is different from person j 's for all $j < i$. Since we can write $B_k = A_k \cap B_{k-1}$, we obtain from equation (C.18) on page 1189 the recurrence

$$\Pr\{B_k\} = \Pr\{B_{k-1}\} \Pr\{A_k \mid B_{k-1}\} ,\tag{5.8}$$

where we take $\Pr\{B_1\} = \Pr\{A_1\} = 1$ as an initial condition. In other words, the probability that $b_1, b_2, \dots, b_k$ are distinct birthdays equals the probability that $b_1, b_2, \dots, b_{k-1}$ are distinct birthdays multiplied by

the probability that $b_k \neq b_i$ for $i = 1, 2, \dots, k-1$, given that $b_1, b_2, \dots, b_{k-1}$ are distinct.

If $b_1, b_2, \dots, b_{k-1}$ are distinct, the conditional probability that $b_k \neq b_i$ for $i = 1, 2, \dots, k-1$ is $\Pr \{A_k \mid B_{k-1}\} = (n - k + 1)/n$, since out of the n days, $n - (k - 1)$ days are not taken. We iteratively apply the recurrence (5.8) to obtain

$$\begin{aligned} \Pr \{B_k\} &= \Pr \{B_{k-1}\} \Pr \{A_k \mid B_{k-1}\} \\ &= \Pr \{B_{k-2}\} \Pr \{A_{k-1} \mid B_{k-2}\} \Pr \{A_k \mid B_{k-1}\} \\ &\vdots \\ &= \Pr \{B_1\} \Pr \{A_2 \mid B_1\} \Pr \{A_3 \mid B_2\} \cdots \Pr \{A_k \mid B_{k-1}\} \\ &= 1 \cdot \left(\frac{n-1}{n}\right) \left(\frac{n-2}{n}\right) \cdots \left(\frac{n-k+1}{n}\right) \\ &= 1 \cdot \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right). \end{aligned}$$

Inequality (3.14) on page 66, $1 + x \leq e^x$, gives us

$$\begin{aligned} \Pr \{B_k\} &\leq e^{-1/n} e^{-2/n} \cdots e^{-(k-1)/n} \\ &= e^{-\sum_{i=1}^{k-1} i/n} \\ &= e^{-k(k-1)/2n} \\ &\leq \frac{1}{2} \end{aligned}$$

when $-k(k-1)/2n \leq \ln(1/2)$. The probability that all k birthdays are distinct is at most $1/2$ when $k(k-1) \geq 2n \ln 2$ or, solving the quadratic equation, when $k \geq (1 + \sqrt{1 + (8 \ln 2)n})/2$. For $n = 365$, we must have $k \geq 23$. Thus, if at least 23 people are in a room, the probability is at least $1/2$ that at least two people have the same birthday. Since a year on Mars is 669 Martian days long, it takes 31 Martians to get the same effect.

An analysis using indicator random variables

Indicator random variables afford a simpler but approximate analysis of the birthday paradox. For each pair (i, j) of the k people in the room,

define the indicator random variable X_{ij} , for $1 \leq i < j \leq k$, by

$$\begin{aligned} X_{ij} &= I\{\text{person } i \text{ and person } j \text{ have the same birthday}\} \\ &= \begin{cases} 1 & \text{if person } i \text{ and person } j \text{ have the same birthday,} \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

By equation (5.7), the probability that two people have matching birthdays is $1/n$, and thus by Lemma 5.1 on page 130, we have

$$\begin{aligned} E[X_{ij}] &= \Pr\{\text{person } i \text{ and person } j \text{ have the same birthday}\} \\ &= 1/n. \end{aligned}$$

Letting X be the random variable that counts the number of pairs of individuals having the same birthday, we have

$$X = \sum_{i=1}^{k-1} \sum_{j=i+1}^k X_{ij}.$$

Taking expectations of both sides and applying linearity of expectation, we obtain

$$\begin{aligned} E[X] &= E\left[\sum_{i=1}^{k-1} \sum_{j=i+1}^k X_{ij}\right] \\ &= \sum_{i=1}^{k-1} \sum_{j=i+1}^k E[X_{ij}] \\ &= \binom{k}{2} \frac{1}{n} \\ &= \frac{k(k-1)}{2n}. \end{aligned}$$

When $k(k-1) \geq 2n$, therefore, the expected number of pairs of people with the same birthday is at least 1. Thus, if we have at least $\sqrt{2n} + 1$ individuals in a room, we can expect at least two to have the same birthday. For $n = 365$, if $k = 28$, the expected number of pairs with the same birthday is $(28 \cdot 27)/(2 \cdot 365) \approx 1.0356$. Thus, with at least 28 people, we expect to find at least one matching pair of birthdays. On Mars, with 669 days per year, we need at least 38 Martians.

The first analysis, which used only probabilities, determined the number of people required for the probability to exceed $1/2$ that a matching pair of birthdays exists, and the second analysis, which used indicator random variables, determined the number such that the expected number of matching birthdays is 1. Although the exact numbers of people differ for the two situations, they are the same asymptotically: $\Theta(\sqrt{n})$.

5.4.2 Balls and bins

Consider a process in which you randomly toss identical balls into b bins, numbered $1, 2, \dots, b$. The tosses are independent, and on each toss the ball is equally likely to end up in any bin. The probability that a tossed ball lands in any given bin is $1/b$. If we view the ball-tossing process as a sequence of Bernoulli trials (see Appendix C.4), where success means that the ball falls in the given bin, then each trial has a probability $1/b$ of success. This model is particularly useful for analyzing hashing (see Chapter 11), and we can answer a variety of interesting questions about the ball-tossing process. (Problem C-2 asks additional questions about balls and bins.)

- *How many balls fall in a given bin?* The number of balls that fall in a given bin follows the binomial distribution $b(k;n, 1/b)$. If you toss n balls, equation (C.41) on page 1199 tells us that the expected number of balls that fall in the given bin is n/b .
- *How many balls must you toss, on the average, until a given bin contains a ball?* The number of tosses until the given bin receives a ball follows the geometric distribution with probability $1/b$ and, by equation (C.36) on page 1197, the expected number of tosses until success is $1/(1/b) = b$.
- *How many balls must you toss until every bin contains at least one ball?* Let us call a toss in which a ball falls into an empty bin a “hit.” We want to know the expected number n of tosses required to get b hits.

Using the hits, we can partition the n tosses into stages. The i th stage consists of the tosses after the $(i - 1)$ st hit up to and

including the i th hit. The first stage consists of the first toss, since you are guaranteed to have a hit when all bins are empty. For each toss during the i th stage, $i - 1$ bins contain balls and $b - i + 1$ bins are empty. Thus, for each toss in the i th stage, the probability of obtaining a hit is $(b - i + 1)/b$.

Let n_i denote the number of tosses in the i th stage. The number of tosses required to get b hits is $n = \sum_{i=1}^b n_i$. Each random variable n_i has a geometric distribution with probability of success $(b - i + 1)/b$ and thus, by equation (C.36), we have

$$\mathbb{E}[n_i] = \frac{b}{b - i + 1} .$$

By linearity of expectation, we have

$$\begin{aligned} \mathbb{E}[n] &= \mathbb{E}\left[\sum_{i=1}^b n_i\right] \\ &= \sum_{i=1}^b \mathbb{E}[n_i] \\ &= \sum_{i=1}^b \frac{b}{b - i + 1} \\ &= b \sum_{i=1}^b \frac{1}{i} \quad (\text{by equation (A.14) on page 1144}) \\ &= b(\ln b + O(1)) \quad (\text{by equation (A.9) on page 1142}) . \end{aligned}$$

It therefore takes approximately $b \ln b$ tosses before we can expect that every bin has a ball. This problem is also known as the *coupon collector's problem*, which says that if you are trying to collect each of b different coupons, then you should expect to acquire approximately $b \ln b$ randomly obtained coupons in order to succeed.

5.4.3 Streaks

Suppose that you flip a fair coin n times. What is the longest streak of consecutive heads that you expect to see? We'll prove upper and lower bounds separately to show that the answer is $\Theta(\lg n)$.

We first prove that the expected length of the longest streak of heads is $O(\lg n)$. The probability that each coin flip is a head is $1/2$. Let A_{ik} be the event that a streak of heads of length at least k begins with the i th coin flip or, more precisely, the event that the k consecutive coin flips $i, i + 1, \dots, i + k - 1$ yield only heads, where $1 \leq k \leq n$ and $1 \leq i \leq n - k + 1$. Since coin flips are mutually independent, for any given event A_{ik} , the probability that all k flips are heads is

$$\Pr \{A_{ik}\} = \frac{1}{2^k} . \tag{5.9}$$

For $k = 2 \lceil \lg n \rceil$,

$$\begin{aligned} \Pr \{A_{i, 2 \lceil \lg n \rceil}\} &= \frac{1}{2^{2 \lceil \lg n \rceil}} \\ &\leq \frac{1}{2^{2 \lg n}} \\ &= \frac{1}{n^2} , \end{aligned}$$

and thus the probability that a streak of heads of length at least $2 \lceil \lg n \rceil$ begins in position i is quite small. There are at most $n - 2 \lceil \lg n \rceil + 1$ positions where such a streak can begin. The probability that a streak of heads of length at least $2 \lceil \lg n \rceil$ begins anywhere is therefore

$$\begin{aligned}
& \Pr \left\{ \bigcup_{i=1}^{n-2\lceil \lg n \rceil + 1} A_{i, 2\lceil \lg n \rceil} \right\} \\
& \leq \sum_{i=1}^{n-2\lceil \lg n \rceil + 1} \Pr \{A_{i, 2\lceil \lg n \rceil}\} \quad (\text{by Boole's inequality (C.21) on page 1190}) \\
& \leq \sum_{i=1}^{n-2\lceil \lg n \rceil + 1} \frac{1}{n^2} \\
& < \sum_{i=1}^n \frac{1}{n^2} \\
& = \frac{1}{n} .
\end{aligned} \tag{5.10}$$

We can use inequality (5.10) to bound the length of the longest streak. For $j = 0, 1, 2, \dots, n$, let L_j be the event that the longest streak of heads has length exactly j , and let L be the length of the longest streak. By the definition of expected value, we have

$$E[L] = \sum_{j=0}^n j \Pr \{L_j\} . \tag{5.11}$$

We could try to evaluate this sum using upper bounds on each $\Pr \{L_j\}$ similar to those computed in inequality (5.10). Unfortunately, this method yields weak bounds. We can use some intuition gained by the above analysis to obtain a good bound, however. For no individual term in the summation in equation (5.11) are both the factors j and $\Pr \{L_j\}$ large. Why? When $j \geq 2\lceil \lg n \rceil$, then $\Pr \{L_j\}$ is very small, and when $j < 2\lceil \lg n \rceil$, then j is fairly small. More precisely, since the events L_j for $j = 0, 1, \dots, n$ are disjoint, the probability that a streak of heads of length at least $2\lceil \lg n \rceil$ begins anywhere is $\sum_{j=2\lceil \lg n \rceil}^n \Pr \{L_j\}$. Inequality (5.10) tells us that the probability that a streak of heads of length at least $2\lceil \lg n \rceil$ begins anywhere is less than $1/n$, which means that $\sum_{j=2\lceil \lg n \rceil}^n \Pr \{L_j\} < 1/n$. Also, noting that $\sum_{j=0}^n \Pr \{L_j\} = 1$, we have that $\sum_{j=0}^{2\lceil \lg n \rceil - 1} \Pr \{L_j\} \leq 1$. Thus, we obtain

$$\begin{aligned}
E[L] &= \sum_{j=0}^n j \Pr\{L_j\} \\
&= \sum_{j=0}^{2\lceil \lg n \rceil - 1} j \Pr\{L_j\} + \sum_{j=2\lceil \lg n \rceil}^n j \Pr\{L_j\} \\
&< \sum_{j=0}^{2\lceil \lg n \rceil - 1} (2\lceil \lg n \rceil) \Pr\{L_j\} + \sum_{j=2\lceil \lg n \rceil}^n n \Pr\{L_j\} \\
&= 2\lceil \lg n \rceil \sum_{j=0}^{2\lceil \lg n \rceil - 1} \Pr\{L_j\} + n \sum_{j=2\lceil \lg n \rceil}^n \Pr\{L_j\} \\
&< 2\lceil \lg n \rceil \cdot 1 + n \cdot \frac{1}{n} \\
&= O(\lg n) .
\end{aligned}$$

The probability that a streak of heads exceeds $r \lceil \lg n \rceil$ flips diminishes quickly with r . Let's get a rough bound on the probability that a streak of at least $r \lceil \lg n \rceil$ heads occurs, for $r \geq 1$. The probability that a streak of at least $r \lceil \lg n \rceil$ heads starts in position i is

$$\begin{aligned}
\Pr\{A_{i,r\lceil \lg n \rceil}\} &= \frac{1}{2^{r\lceil \lg n \rceil}} \\
&\leq \frac{1}{n^r} .
\end{aligned}$$

A streak of at least $r \lceil \lg n \rceil$ heads cannot start in the last $n - r \lceil \lg n \rceil + 1$ flips, but let's overestimate the probability of such a streak by allowing it to start anywhere within the n coin flips. Then the probability that a streak of at least $r \lceil \lg n \rceil$ heads occurs is at most

$$\begin{aligned}
\Pr\left\{\bigcup_{i=1}^n A_{i,r\lceil \lg n \rceil}\right\} &\leq \sum_{i=1}^n \Pr\{A_{i,r\lceil \lg n \rceil}\} \quad (\text{by Boole's inequality (C.21)}) \\
&\leq \sum_{i=1}^n \frac{1}{n^r} \\
&= \frac{1}{n^{r-1}} .
\end{aligned}$$

Equivalently, the probability is at least $1 - 1/n^{r-1}$ that the longest streak has length less than $r \lceil \lg n \rceil$.

As an example, during $n = 1000$ coin flips, the probability of encountering a streak of at least $2 \lceil \lg n \rceil = 20$ heads is at most $1/n = 1/1000$. The chance of a streak of at least $3 \lceil \lg n \rceil = 30$ heads is at most $1/n^2 = 1/1,000,000$.

Let's now prove a complementary lower bound: the expected length of the longest streak of heads in n coin flips is $\Omega(\lg n)$. To prove this bound, we look for streaks of length s by partitioning the n flips into approximately n/s groups of s flips each. If we choose $s = \lfloor (\lg n)/2 \rfloor$, we'll see that it is likely that at least one of these groups comes up all heads, which means that it's likely that the longest streak has length at least $s = \Omega(\lg n)$. We'll then show that the longest streak has expected length $\Omega(\lg n)$.

Let's partition the n coin flips into at least $\lfloor n / \lfloor (\lg n)/2 \rfloor \rfloor$ groups of $\lfloor (\lg n)/2 \rfloor$ consecutive flips and bound the probability that no group comes up all heads. By equation (5.9), the probability that the group starting in position i comes up all heads is

$$\begin{aligned} \Pr \{A_{i, \lfloor (\lg n)/2 \rfloor}\} &= \frac{1}{2^{\lfloor (\lg n)/2 \rfloor}} \\ &\geq \frac{1}{\sqrt{n}}. \end{aligned}$$

The probability that a streak of heads of length at least $\lfloor (\lg n)/2 \rfloor$ does not begin in position i is therefore at most $1 - 1/\sqrt{n}$. Since the $\lfloor n / \lfloor (\lg n)/2 \rfloor \rfloor$ groups are formed from mutually exclusive, independent coin flips, the probability that every one of these groups *fails* to be a streak of length $\lfloor (\lg n)/2 \rfloor$ is at most

$$\begin{aligned} (1 - 1/\sqrt{n})^{\lfloor n / \lfloor (\lg n)/2 \rfloor \rfloor} &\leq (1 - 1/\sqrt{n})^{n / \lfloor (\lg n)/2 \rfloor - 1} \\ &\leq (1 - 1/\sqrt{n})^{2n / \lg n - 1} \\ &\leq e^{-(2n / \lg n - 1) / \sqrt{n}} \\ &= O(e^{-\ln n}) \\ &= O(1/n). \end{aligned} \tag{5.12}$$

For this argument, we used inequality (3.14), $1 + x \leq e^x$, on page 66 and the fact, which you may verify, that $(2n/\lg n - 1)/\sqrt{n} \geq \ln n$ for sufficiently large n .

We want to bound the probability that the longest streak equals or exceeds $\lfloor (\lg n)/2 \rfloor$. To do so, let L be the event that the longest streak of heads equals or exceeds $s = \lfloor (\lg n)/2 \rfloor$. Let $\bar{L}$ be the complementary event, that the longest streak of heads is strictly less than s , so that $\Pr \{L\} + \Pr \{\bar{L}\} = 1$. Let F be the event that every group of s flips fails to be a streak of s heads. By inequality (5.12), we have $\Pr \{F\} = O(1/n)$. If the longest streak of heads is less than s , then certainly every group of s flips fails to be a streak of s heads, which means that event $\bar{L}$ implies event F . Of course, event F could occur even if event $\bar{L}$ does not (for example, if a streak of s or more heads crosses over the boundary between two groups), and so we have $\Pr \{\bar{L}\} \leq \Pr \{F\} = O(1/n)$. Since $\Pr \{L\} + \Pr \{\bar{L}\} = 1$, we have that

$$\begin{aligned} \Pr \{L\} &= 1 - \Pr \{\bar{L}\} \\ &\geq 1 - \Pr \{F\} \\ &= 1 - O(1/n). \end{aligned}$$

That is, the probability that the longest streak equals or exceeds $\lfloor (\lg n)/2 \rfloor$ is

$$\sum_{j=\lfloor (\lg n)/2 \rfloor}^n \Pr \{L_j\} \geq 1 - O(1/n). \quad (5.13)$$

We can now calculate a lower bound on the expected length of the longest streak, beginning with equation (5.11) and proceeding in a manner similar to our analysis of the upper bound:

$$\begin{aligned}
E[L] &= \sum_{j=0}^n j \Pr\{L_j\} \\
&= \sum_{j=0}^{\lfloor (\lg n)/2 \rfloor - 1} j \Pr\{L_j\} + \sum_{j=\lfloor (\lg n)/2 \rfloor}^n j \Pr\{L_j\} \\
&\geq \sum_{j=0}^{\lfloor (\lg n)/2 \rfloor - 1} 0 \cdot \Pr\{L_j\} + \sum_{j=\lfloor (\lg n)/2 \rfloor}^n \lfloor (\lg n)/2 \rfloor \Pr\{L_j\} \\
&= 0 \cdot \sum_{j=0}^{\lfloor (\lg n)/2 \rfloor - 1} \Pr\{L_j\} + \lfloor (\lg n)/2 \rfloor \sum_{j=\lfloor (\lg n)/2 \rfloor}^n \Pr\{L_j\} \\
&\geq 0 + \lfloor (\lg n)/2 \rfloor (1 - O(1/n)) \quad (\text{by inequality (5.13)}) \\
&= \Omega(\lg n) .
\end{aligned}$$

As with the birthday paradox, we can obtain a simpler, but approximate, analysis using indicator random variables. Instead of determining the expected length of the longest streak, we'll find the expected number of streaks with at least a given length. Let $X_{ik} = I_{\{A_{ik}\}}$ be the indicator random variable associated with a streak of heads of length at least k beginning with the i th coin flip. To count the total number of such streaks, define

$$X_k = \sum_{i=1}^{n-k+1} X_{ik} .$$

Taking expectations and using linearity of expectation, we have

$$\begin{aligned}
E[X_k] &= E\left[\sum_{i=1}^{n-k+1} X_{ik}\right] \\
&= \sum_{i=1}^{n-k+1} E[X_{ik}] \\
&= \sum_{i=1}^{n-k+1} \Pr\{A_{ik}\} \\
&= \sum_{i=1}^{n-k+1} \frac{1}{2^k} \\
&= \frac{n-k+1}{2^k}.
\end{aligned}$$

By plugging in various values for k , we can calculate the expected number of streaks of length at least k . If this expected number is large (much greater than 1), then we expect many streaks of length k to occur, and the probability that one occurs is high. If this expected number is small (much less than 1), then we expect to see few streaks of length k , and the probability that one occurs is low. If $k = c \lg n$, for some positive constant c , we obtain

$$\begin{aligned}
E[X_{c \lg n}] &= \frac{n - c \lg n + 1}{2^{c \lg n}} \\
&= \frac{n - c \lg n + 1}{n^c} \\
&= \frac{1}{n^{c-1}} - \frac{(c \lg n - 1)/n}{n^{c-1}} \\
&= \Theta(1/n^{c-1}).
\end{aligned}$$

If c is large, the expected number of streaks of length $c \lg n$ is small, and we conclude that they are unlikely to occur. On the other hand, if $c = 1/2$, then we obtain $E[X_{(1/2) \lg n}] = \Theta(1/n^{1/2-1}) = \Theta(n^{1/2})$, and we expect there to be numerous streaks of length $(1/2) \lg n$. Therefore, one streak of such a length is likely to occur. We can conclude that the expected length of the longest streak is $\Theta(\lg n)$.

5.4.4 The online hiring problem

As a final example, let's consider a variant of the hiring problem. Suppose now that you do not wish to interview all the candidates in order to find the best one. You also want to avoid hiring and firing as you find better and better applicants. Instead, you are willing to settle for a candidate who is close to the best, in exchange for hiring exactly once. You must obey one company requirement: after each interview you must either immediately offer the position to the applicant or immediately reject the applicant. What is the trade-off between minimizing the amount of interviewing and maximizing the quality of the candidate hired?

We can model this problem in the following way. After meeting an applicant, you are able to give each one a score. Let $score(i)$ denote the score you give to the i th applicant, and assume that no two applicants receive the same score. After you have seen j applicants, you know which of the j has the highest score, but you do not know whether any of the remaining $n - j$ applicants will receive a higher score. You decide to adopt the strategy of selecting a positive integer $k < n$, interviewing and then rejecting the first k applicants, and hiring the first applicant thereafter who has a higher score than all preceding applicants. If it turns out that the best-qualified applicant was among the first k interviewed, then you hire the n th applicant—the last one interviewed. We formalize this strategy in the procedure **ONLINE-MAXIMUM**(k, n), which returns the index of the candidate you wish to hire.

ONLINE-MAXIMUM(k, n)

```
1 best-score =  $-\infty$ 
2 for  $i = 1$  to  $k$ 
3     if  $score(i) > best\text{-}score$ 
4          $best\text{-}score = score(i)$ 
5 for  $i = k + 1$  to  $n$ 
6     if  $score(i) > best\text{-}score$ 
7         return  $i$ 
8 return  $n$ 
```

If we determine, for each possible value of k , the probability that you hire the most qualified applicant, then you can choose the best possible k and implement the strategy with that value. For the moment, assume that k is fixed. Let $M(j) = \max \{score(i) : 1 \leq i \leq j\}$ denote the maximum score among applicants 1 through j . Let S be the event that you succeed in choosing the best-qualified applicant, and let S_i be the event that you succeed when the best-qualified applicant is the i th one interviewed. Since the various S_i are disjoint, we have that $\Pr\{S\} = \sum_{i=1}^n \Pr\{S_i\}$. Noting that you never succeed when the best-qualified applicant is one of the first k , we have that $\Pr\{S_i\} = 0$ for $i = 1, 2, \dots, k$. Thus, we obtain

$$\Pr\{S\} = \sum_{i=k+1}^n \Pr\{S_i\} . \quad (5.14)$$

We now compute $\Pr\{S_i\}$. In order to succeed when the best-qualified applicant is the i th one, two things must happen. First, the best-qualified applicant must be in position i , an event which we denote by B_i . Second, the algorithm must not select any of the applicants in positions $k + 1$ through $i - 1$, which happens only if, for each j such that $k + 1 \leq j \leq i - 1$, line 6 finds that $score(j) < best-score$. (Because scores are unique, we can ignore the possibility of $score(j) = best-score$.) In other words, all of the values $score(k + 1)$ through $score(i - 1)$ must be less than $M(k)$. If any are greater than $M(k)$, the algorithm instead returns the index of the first one that is greater. We use O_i to denote the event that none of the applicants in position $k + 1$ through $i - 1$ are chosen. Fortunately, the two events B_i and O_i are independent. The event O_i depends only on the relative ordering of the values in positions 1 through $i - 1$, whereas B_i depends only on whether the value in position i is greater than the values in all other positions. The ordering of the values in positions 1 through $i - 1$ does not affect whether the value in position i is greater than all of them, and the value in position i does not affect the ordering of the values in positions 1 through $i - 1$. Thus, we can apply equation (C.17) on page 1188 to obtain

$$\Pr \{S_i\} = \Pr \{B_i \cap O_i\} = \Pr \{B_i\} \Pr \{O_i\}.$$

We have $\Pr \{B_i\} = 1/n$ since the maximum is equally likely to be in any one of the n positions. For event O_i to occur, the maximum value in positions 1 through $i-1$, which is equally likely to be in any of these $i-1$ positions, must be in one of the first k positions. Consequently, $\Pr \{O_i\} = k/(i-1)$ and $\Pr \{S_i\} = k/(n(i-1))$. Using equation (5.14), we have

$$\begin{aligned} \Pr \{S\} &= \sum_{i=k+1}^n \Pr \{S_i\} \\ &= \sum_{i=k+1}^n \frac{k}{n(i-1)} \\ &= \frac{k}{n} \sum_{i=k+1}^n \frac{1}{i-1} \\ &= \frac{k}{n} \sum_{i=k}^{n-1} \frac{1}{i}. \end{aligned}$$

We approximate by integrals to bound this summation from above and below. By the inequalities (A.19) on page 1150, we have

$$\int_k^n \frac{1}{x} dx \leq \sum_{i=k}^{n-1} \frac{1}{i} \leq \int_{k-1}^{n-1} \frac{1}{x} dx.$$

Evaluating these definite integrals gives us the bounds

$$\frac{k}{n}(\ln n - \ln k) \leq \Pr \{S\} \leq \frac{k}{n}(\ln(n-1) - \ln(k-1)),$$

which provide a rather tight bound for $\Pr \{S\}$. Because you wish to maximize your probability of success, let us focus on choosing the value of k that maximizes the lower bound on $\Pr \{S\}$. (Besides, the lower-bound expression is easier to maximize than the upper-bound expression.) Differentiating the expression $(k/n)(\ln n - \ln k)$ with respect to k , we obtain

$$\frac{1}{n}(\ln n - \ln k - 1).$$

Setting this derivative equal to 0, we see that you maximize the lower bound on the probability when $\ln k = \ln n - 1 = \ln(n/e)$ or, equivalently, when $k = n/e$. Thus, if you implement our strategy with $k = n/e$, you succeed in hiring the best-qualified applicant with probability at least $1/e$.

Exercises

5.4-1

How many people must there be in a room before the probability that someone has the same birthday as you do is at least $1/2$? How many people must there be before the probability that at least two people have a birthday on July 4 is greater than $1/2$?

5.4-2

How many people must there be in a room before the probability that two people have the same birthday is at least 0.99? For that many people, what is the expected number of pairs of people who have the same birthday?

5.4-3

You toss balls into b bins until some bin contains two balls. Each toss is independent, and each ball is equally likely to end up in any bin. What is the expected number of ball tosses?

★ 5.4-4

For the analysis of the birthday paradox, is it important that the birthdays be mutually independent, or is pairwise independence sufficient? Justify your answer.

★ 5.4-5

How many people should be invited to a party in order to make it likely that there are *three* people with the same birthday?

★ 5.4-6

What is the probability that a k -string (defined on page 1179) over a set of size n forms a k -permutation? How does this question relate to the

birthday paradox?

★ 5.4-7

You toss n balls into n bins, where each toss is independent and the ball is equally likely to end up in any bin. What is the expected number of empty bins? What is the expected number of bins with exactly one ball?

★ 5.4-8

Sharpen the lower bound on streak length by showing that in n flips of a fair coin, the probability is at least $1 - 1/n$ that a streak of length $\lg n - 2 \lg \lg n$ consecutive heads occurs.

Problems

5-1 Probabilistic counting

With a b -bit counter, we can ordinarily only count up to $2^b - 1$. With R. Morris's *probabilistic counting*, we can count up to a much larger value at the expense of some loss of precision.

We let a counter value of i represent a count of n_i for $i = 0, 1, \dots, 2^b - 1$, where the n_i form an increasing sequence of nonnegative values. We assume that the initial value of the counter is 0, representing a count of $n_0 = 0$. The INCREMENT operation works on a counter containing the value i in a probabilistic manner. If $i = 2^b - 1$, then the operation reports an overflow error. Otherwise, the INCREMENT operation increases the counter by 1 with probability $1/(n_{i+1} - n_i)$, and it leaves the counter unchanged with probability $1 - 1/(n_{i+1} - n_i)$.

If we select $n_i = i$ for all $i \geq 0$, then the counter is an ordinary one. More interesting situations arise if we select, say, $n_i = 2^{i-1}$ for $i > 0$ or $n_i = F_i$ (the i th Fibonacci number—see equation (3.31) on page 69).

For this problem, assume that n_{2^b-1} is large enough that the probability of an overflow error is negligible.